

1) What is the difference between slice() and splice()?

slice():-

Slice return new array from the main array without any changes

Syntax:-

slice(start index,end index);

Example:-

Let arr=[10,20,30,40,50]

Let result=arr.slice(2,4);

console.log(result);

Output:-

[30,40,50]

Splice:-

Splice is used to add elements, remove elements from main array and returns modified array

Syntax:-

splice(start index,deleted count,item1,item2...);

Example:-

let arr = [10, 20, 30, 40, 50];

let result = arr.splice(1, 2);

console.log(result);

console.log(arr);

Output:-

[20, 30]

[10, 40, 50]

2)

DOM (Document Object Model) is used to manipulate and control HTML elements of a web page. Its main object is document.

example:-<p id="demo">Hello</p>

<button onclick="changeText()">Click Me</button>

BOM (Browser Object Model) is used to interact with the browser, such as navigation, alerts, and history. Its main object is window.

example:-window.alert("Hello User!");

3)closure is nothing but it remember the outer function variables even after the function execution is completed

Example:-

```
function outer() {  
    let count = 0;  
  
    return function inner() {  
        count++;  
        console.log(count);  
    }  
}
```

```
let result = outer();
```

```
result(); // 1
```

```
result(); // 2
```

```
result(); //3
```

4)Event propagation is the process by which an event travels through the DOM elements in two phases: capturing (top to bottom) and bubbling (bottom to top).

Capturing:-

HTML → BODY → DIV → BUTTON

Bubbling

BUTTON → DIV → BODY → HTML

5)promise is used to handle asynchronous operations and give results in future (success or failure) Promise is an object that represents whether an asynchronous operation is:

- Pending
- Fulfilled (Success)
- Rejected (Failure)

It is used to perform asynchronous operations.

```
6)let arr = [1, 2, 2, 3, 4, 4, 5];
```

```
let unique = arr.filter((value, index) => {  
  return arr.indexOf(value) === index;  
});
```

```
console.log(unique);
```

Output:-

```
[1,2,3,4,5]
```

indexOf():- indexOf returns the first index of a given element, or -1 if not found.

```
7)let matrix = [  
  [1, 2, 3],  
  [4, 5, 6]  
];
```

```
let transpose = [];
```

```
for (let i = 0; i < matrix[0].length; i++) {  
  transpose[i] = [];
```

```
  for (let j = 0; j < matrix.length; j++) {  
    transpose[i][j] = matrix[j][i];  
  }  
}
```

```
console.log(transpose);
```

```
8)
```